Alexander Nicholas Sietsema

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Research Interests

Mathematical Data Science, Machine Learning, Numerical Linear Algebra, Optimization, Applications.

Citizenship

USA

Education

Ph.D., Computational and Applied Mathematics (in progress) 2022 – 2027 (proj.)
University of California, Los Angeles, Advisor: Deanna Needell Los Angeles, CA
Advanced to candidacy Fall 2024

M.A., Computational and Applied Mathematics 2022 – 2024
University of California, Los Angeles Los Angeles, CA

B.S., Advanced Mathematics; B.S., Computational Mathematics 2018 – 2022
Michigan State University East Lansing, MI

Publications

* indicates equal contribution

Journal Publications

4. Benjamin Jarman, Lara Kassab, Deanna Needell, **Alexander Sietsema**. "Stochastic Iterative Methods for Online Rank Aggregation from Pairwise Comparisons". *BIT Numerical Mathematics*, vol. 64, p. 26 (2024). https://link.springer.com/article/10.1007/s10543-024-01024-x

3. Rachel Domagalski, Sergi Elizalde, Jinting Liang, Quinn Minnich, Bruce E. Sagan, Jamie Schmidt, **Alexander Sietsema**. "Cyclic Pattern Containment and Avoidance". *Advances in Applied Mathematics*, vol. 135, p. 102320 (2022). https://www.sciencedirect.com/science/article/abs/pii/S019688582200001X

2. Rachel Domagalski, Jinting Liang, Quinn Minnich, Bruce E. Sagan, Jamie Schmidt, **Alexander Sietsema**. "Pinnacle Set Properties". *Discrete Mathematics*, vol. 345, iss. 7, p. 112882 (2022). https://www.sciencedirect.com/science/article/abs/pii/S0012365X22000887

1. Rachel Domagalski, Jinting Liang, Quinn Minnich, Bruce E. Sagan, Jamie Schmidt, **Alexander Sietsema**. "Cyclic Shuffle Compatibility". *Séminaire Lotharingien de Combinatoire*, vol. 85d (2021). https://www.mat.univie.ac.at/~slc/wpapers/s85domasaga.pdf

Conference Publications

6. **Alexander Sietsema**, Deanna Needell, Yaniv Plan. "Binarization in low-rank approximation and non-negative matrix factorization". *To appear, Proc. 60th Asilomar Conf. on Signals, Systems and Computers* (2026).

5. Alejandra Castillo, Jamie Haddock, Iryna Hartsock, Lara Kassab, Deanna Needell, **Alexander Sietsema**. "Geometry-aware Randomized Kaczmarz: fairer solutions on non-uniform geometries". *To appear, Proc. 60th Asilomar Conf. on Signals, Systems and Computers* (2026).

4. Kedar Karhadkar*, **Alexander Sietsema***, Deanna Needell, Guido Montúfar. "Harmful Overfitting in Sobolev Spaces". *Forty-Third International Conference on Machine Learning (ICML)* (2026). https://arxiv.org/abs/2602.00825

3. David R. Johnson, **Alexander Sietsema**, Rishabh Anand, Deanna Needell, Smita Krishnaswamy, Michael Perlmutter. "VDW-GNNs: Vector diffusion wavelets for geometric graph neural networks". *Forty-Third International Conference on Machine Learning (ICML)* (2026). https://arxiv.org/abs/2510.01022

2. **Alexander Sietsema***, Zerrin Vural*, James Chapman, Yotam Yaniv, Deanna Needell. "Stratified Non-Negative Tensor Factorization". *Proc. 58th Asilomar Conf. on Signals, Systems and Computers*. IEEE, pp. 260–264 (2024). https://ieeexplore.ieee.org/document/10942969

1. **Alexander N. Sietsema**, Michael T. McCann, Marc L. Klasky, Saiprasad Ravishankar. "Comparing One-step and Two-step Scatter Correction And Density Reconstruction In X-Ray CT". *7th International Conference on Image Formation in X-Ray Computed Tomography.* SPIE, vol. 12304, pp. 532–538 (2022). `https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12304/2647151/Comparing-one-step-and-two-step-scatter-correction-and-density/10.1117/12.2647151`

Teaching Experience

Python With Applications II (PIC 16B) Teaching Assistant Spring 2023 – Spring 2026
Wrote discussion materials, led discussion sessions, evaluated student projects.

Python With Applications I (PIC 16A) Teaching Assistant Fall 2022, Winter 2023, Spring 2024
Wrote discussion materials, led discussion sessions, graded exams, led study sessions.

Honors Linear Algebra (MTH 317H) Undergraduate Learning Assistant Fall 2021
Led recitation sessions, graded homeworks and exams, led study sessions, held LaTeX learning sessions.

Calculus I (MTH 132) Course Assistant Spring 2020
Answered questions on Piazza, led biweekly help sessions for students, graded exams.

Calculus II (MTH 133) Undergraduate Learning Assistant Fall 2019
Supervised two sections, led recitation sessions, led special review sessions, graded labs, quizzes, and exams.

Presentations

Conference Oral Presentations

4. **"Harmful Overfitting in Sobolev Spaces"**, SIAM Conference on Imaging Science, Salt Lake City, UT, November 2026 (scheduled).
3. **"A Geometric Approach to Overfitting"**, The 9th International Conference on Computational Harmonic Analysis *(Invited)*, Nashville, TN, May 2026.
2. **"Harmful Overfitting in Sobolev Spaces"**, Southern California Applied Mathematics Symposium, Los Angeles, CA, April 2026.
1. **"Stratified Non-Negative Tensor Factorization"**, 58th Asilomar Conf. on Signals, Systems, and Computers, Pacific Grove, CA, Oct. 2024.

Seminar Talks

4. **"A Geometric Approach to Overfitting"**, Codes and Expansions (CodEx) Seminar, Online, August 2026.
3. **"Harmful Overfitting in Sobolev Spaces"**, Michigan State University Applied Mathematics Seminar *(Invited)*, East Lansing, MI, April 2026.
2. **"A Stochastic Subtraction Game"**, Michigan State University Graduate and Undergraduate Student Seminar, East Lansing, MI, May 2022.
1. **"Pattern Avoidance in Cyclic Permutations"**, Michigan State University Graduate and Undergraduate Student Seminar, East Lansing, MI, Jan. 2021.

Posters

18. **"Binarization in low-rank approximation and non-negative matrix factorization"**, 60th Asilomar Conf. on Signals, Systems, and Computers, Pacific Grove, CA, Oct. 2026 (scheduled).
17. **"Geometry-aware Randomized Kaczmarz: fairer solutions on non-uniform geometries"**, 60th Asilomar Conf. on Signals, Systems, and Computers, Pacific Grove, CA, Oct. 2026 (scheduled).
16. **"Binarization in Non-Negative Matrix Factorization"**, Foundations of Computational Mathematics, Vienna, Austria, July 2026.
15. **"Harmful Overfitting in Sobolev Spaces"**, Foundations of Computational Mathematics, Vienna, Austria, July 2026.
14. **"VDW-GNNs: Vector diffusion wavelets for geometric graph neural networks"**, ICML, Seoul, South Korea, July 2026.
13. **"Harmful Overfitting in Sobolev Spaces"**, ICML, Seoul, South Korea, July 2026.
12. **"Harmful Overfitting in Sobolev Spaces"**, ICTP-INdAM-SLMath Summer Graduate School for Machine Learning, Trieste, Italy, June 2026.

11. **"VDW-GNNs: Vector diffusion wavelets for geometric graph neural networks"**, IPAM Workshop: Mathematics of Cancer: Open Mathematical Problems, Los Angeles, CA, Feb. 2026.
10. **"Stochastic Iterative Methods for Online Rank Aggregation from Pairwise Comparisons"**, 2nd Conf. on Random Matrix Theory and Numerical Linear Algebra, Seattle, WA, June 2025.
9. **"Stratified Non-Negative Tensor Factorization"**, Workshop: Approximation And Learning In High Dimensions, Centre de Recherches Mathématiques, Montreal, QC, Canada, June 2025.
8. **"Stochastic Iterative Methods for Online Rank Aggregation from Pairwise Comparisons"**, Research in the Age of AI Symposium, Los Angeles, CA, Feb. 2024.
7. **"Comparing One-Step and Two-Step Descattering and Reconstruction"**, CT Meeting 2022, Baltimore, MD, June 2022.
6. **"Comparing One-Step and Two-Step Descattering and Reconstruction"**, MSU CMSE Student Research Symposium, East Lansing, MI, May 2022.
5. **"An Algorithm For Counting Admissible Pinnacle Orderings"**, Permutation Patterns 2021 (Univ. of Strathclyde Combinatorics Group), June 2021.
4. **"Pattern Avoidance in Cyclic Permutations"**, Joint Mathematics Meetings Poster Session, Jan. 2021.
3. **"A Cyclic Variant of the Erdős-Szekeres Theorem"**, Joint Mathematics Meetings Poster Session, Jan. 2021.
2. **"Pattern Avoidance in Cyclic Permutations"**, JMU SUMS Poster Session, Dec. 2020.
1. **"A Cyclic Variant of the Erdős-Szekeres Theorem"**, JMU SUMS Poster Session, Dec. 2020.

Honors and Awards

Carolyn D. Smith Graduate Scholarship ($5,477) — 2026
UCLA Scholarship

Doctoral Travel Grant ($1,000) — 2026
UCLA Graduate Division Award

Dissertation Year Fellowship Award ($32,000) — 2026
UCLA Department of Mathematics Award

Liggett Departmental Teaching Assistant Award ($500) — 2026
UCLA Department of Mathematics Teaching Award

Featured Student, Graduate Student Spotlight — 2026
UCLA Department of Mathematics

RMT NLA II Travel Award ($800) — 2025
2nd Conf. on Random Matrix Theory and Numerical Linear Algebra Award

CRM Thematic Program Travel Award ($1,300) — 2025
Centre de Recherches Mathématiques Award

Outstanding Poster — 2021
Joint Mathematics Meetings Poster Session, "Pattern Avoidance in Cyclic Permutations"

Honorable Mention Poster — 2021
Joint Mathematics Meetings Poster Session, "A Cyclic Variant of the Erdős-Szekeres Theorem"

Herbert T. Graham Scholarship ($6,000) — 2020, 2021, 2022
Michigan State University Department of Mathematics Award

Paul and Wilma Dressel Endowed Scholarship ($2,000) — 2019
Michigan State University Department of Mathematics Award

FAITH Endowment Scholarship for Academic Excellence ($15,500) — 2018 – 2022
FAITH: Endowment for Greek Orthodoxy and Hellenism

Michigan State University Alumni Distinguished Freshman ($61,022) — 2018 – 2022
Michigan State University Full-Tuition Scholarship

Dean's List — 2018 – 2022
(all undergraduate semesters)

Additional Selected Projects

Scripps National Spelling Bee — Spring 2024
(Subject to NDA)
Analysis of word list difficulty leveling and in-competition word selection.

Honors Senior Thesis — Spring 2022
Advisor: Albert Cohen
Exploring game theoretic properties and theorems for a novel stochastic variant of the classical subtraction game, including optimal move selection and conditions for excluding available moves, with applications to sports analytics.

Projects in Industrial Mathematics Spring 2022
Advisor: Peiru Wu
Creating a data handling pipeline for hospital Medicare and Medicaid cost reports, as well as investigating trends in those reports. Industry project with The Rybar Group.

MSU Risk Management and Sports Analytics Group Fall 2021
Advisor: Albert Cohen
Developing new methods for optimal decision making for two-point conversion attempts in American football; analyzing the effects of fights in hockey on the outcomes of games.

UCLA Computational and Applied Mathematics REU Summer 2021
Advisor: Jamie Haddock
Exploring Kaczmarz methods for inconsistent and corrupted linear systems and their connections to maximum likelihood estimation techniques for ranking sports teams.